AERO40009 Thermodynamics
Tutorial Sheet 2

Consider questions 1 & 2 as a pair.

Question 1.

1. One kg of air undergoes a slow fully resisted adiabatic process during which at all times Pv^{γ} = constant, where $C_p/C_V = \gamma = 1.4$.

Initially $P = 1$ bar, T = 300K. Finally $P = 10$ bar

(a) What is the final temperature? (579K)
(b) What is the increase of energy? (201kJ)
(c) What is the work transfer?

Question 2.

One kg of air expands from 10 bar, 579k to a pressure of 1 bar. The process is adiabatic, but irreversible, and it is found that initial and final states are related by

$Pv^{1.25}$ = constant.

(a) What is the final temperature? (365.3K)
(b) What is the change of energy? (-153kJ)
(c) What is the work transfer?

Note: the equation $Pv^{1.25}$ = constant in the above question describes a particular process. It is not the case that γ has changed value - the ratio of specific heats, γ, has a value of 1.4 for air, irrespective of the process.

1. (a) $\because Pv^{\gamma} = \text{constant}$

$\therefore v_I^{1.4} = 10\, v_F^{1.4}$ $\quad\therefore v_I = 10^{\frac{5}{7}} v_F$

$\because Pv = RT$ $\quad\therefore v_I = 300R,\ 10 v_F = T_F R$

$\therefore \dfrac{T_F R}{300R} = \dfrac{10 v_F}{v_I} = 10^{-\frac{5}{7}}$

$\therefore T_F = 300 \times 10 \times 10^{-\frac{5}{7}} = 579.21 \ (K)$

(b) $\Delta E = \Delta E_{cv}$

$$= M_F C_v T_F - M_I C_v T_I = \frac{MC_p}{\gamma}(T_F - T_I)$$

$$= 200.43 kJ$$

(c) $\because Q - W = \Delta E$

$\therefore W = -\Delta E = -200.43 kJ$

2. (a) From 1(a), $T_F = 579 \times \frac{1}{10} \times 10^{\frac{4}{5}} = 365.32 \ (K)$

(b) From 1(b),

$$\Delta E = \frac{MC_p}{\gamma}(T_F - T_I) = -153.39 kJ$$

(c) From 1(c),

$$W = -\Delta E = 153.39 kJ$$

Control Volume Analysis

Question 3.

An insulated rigid container of volume V contains a perfect gas at temperature T_1, and pressure P_1. It is connected to a pipe containing the same gas at constant temperature T_0 and pressure P_0. Gas enters the container through a small valve until the pressure within the container is P_2. Using mass conservation and the first law statement:

(a) Show that

$$\frac{P_2}{T_2} - \frac{P_1}{T_1} = \frac{P_2 - P_1}{\gamma T_0}$$

(b) Show that the heat transfer which would then be required to bring the contents to a final temperature T_0 is

$$\frac{p_2 V}{\gamma - 1}\left[\frac{p_1}{p_2}\left(\frac{T_0}{T_1} - \frac{1}{\gamma}\right) - \left(1 - \frac{1}{\gamma}\right)\right].$$

(a) $M_{in} = M_2 - M_1 = \dfrac{P_2 V_2}{R T_2} - \dfrac{P_1 V_1}{R T_1}$

$\because V = V_1 = V_2 \quad \therefore M_{in} = \left(\dfrac{P_2}{T_2} - \dfrac{P_1}{T_1}\right)\dfrac{V}{R}$

$Q_{12} - W_x = \Delta E_{cv} + \Sigma \delta m_{out} h_{o,out} - \Sigma \delta m_{in} h_{o,in}$

$\therefore 0 - 0 = M_2 C_v T_2 - M_1 C_v T_1 - M_{in} C_p T_{in}$

$\therefore M_1 C_v T_1 - M_2 C_v T_2 = M_{in} \gamma C_v T_0 = \left(\dfrac{P_2}{T_2} - \dfrac{P_1}{T_1}\right)\dfrac{V}{R}\gamma C_v T_0$

$\therefore M_1 T_1 - M_2 T_2 = \left(\dfrac{P_2}{T_2} - \dfrac{P_1}{T_1}\right)\dfrac{V}{R}\gamma T_0$

$\therefore \dfrac{P_1 V_1}{R T_1} T_1 - \dfrac{P_2 V_2}{R T_2} T_2 = (P_1 - P_2)\dfrac{V}{R} = \left(\dfrac{P_2}{T_2} - \dfrac{P_1}{T_1}\right)\dfrac{V}{R}\gamma T_0$

$\therefore \dfrac{P_2}{T_2} - \dfrac{P_1}{T_1} = \dfrac{P_1 - P_2}{\gamma T_0}$

Question 3.

An insulated rigid container of volume V contains a perfect gas at temperature T_1, and pressure P_1. It is connected to a pipe containing the same gas at constant temperature T_0 and pressure P_0. Gas enters the container through a small valve until the pressure within the container is P_2. Using mass conservation and the first law statement:

(a) Show that

$$\frac{P_2}{T_2} - \frac{P_1}{T_1} = \frac{P_2 - P_1}{\gamma T_0}$$

(b) Show that the heat transfer which would then be required to bring the contents to a final temperature T_0 is

$$\frac{p_2 V}{\gamma - 1}\left[\frac{p_1}{p_2}\left(\frac{T_0}{T_1} - \frac{1}{\gamma}\right) - \left(1 - \frac{1}{\gamma}\right)\right].$$

(b) $T_2 \to T_0$, V , M_2

$$Q = W_x + \Delta E_{cv} + \Sigma \delta m_{out} h_{o,out} - \Sigma \delta m_{in} h_{o,in}$$

$$= \Delta E_{cv} = M_2 C_v T_0 - M_2 C_v T_2$$

From (a), $\frac{P_2 T_1 - P_1 T_2}{T_1 T_2} = \frac{P_2 - P_1}{\gamma T_0}$ $\quad \therefore T_2 = \frac{(P_2 T_1 - P_1 T_2)\gamma T_0}{(P_2 - P_1) T_1}$

$\therefore Q =$

$\because R = C_p - C_v = C_p - \frac{C_p}{\gamma} = \gamma C_v - C_v \quad \therefore \frac{R}{C_v} = \gamma - 1$

$$\therefore Q = \frac{P_2 V_2}{R T_2} C_v (T_0 - T_2) = \frac{P_2 V_2}{(\gamma - 1) T_2}(T_0 - T_2) = \frac{P_2 V_2}{\gamma - 1}\left(\frac{T_0}{T_2} - 1\right)$$

$\because \frac{P_2}{T_2} - \frac{P_1}{T_1} = \frac{P_2 - P_1}{\gamma T_0} \quad \therefore \frac{T_0}{T_2} P_2 - \frac{T_0}{T_1} P_1 = \frac{P_2 - P_1}{\gamma}$

Question 3.

An insulated rigid container of volume V contains a perfect gas at temperature T_1, and pressure P_1. It is connected to a pipe containing the same gas at constant temperature T_0 and pressure P_0. Gas enters the container through a small valve until the pressure within the container is P_2. Using mass conservation and the first law statement:

(a) Show that

$$\frac{P_2}{T_2} - \frac{P_1}{T_1} = \frac{P_2 - P_1}{\gamma T_0}$$

(b) Show that the heat transfer which would then be required to bring the contents to a final temperature T_0 is

$$\frac{p_2 V}{\gamma - 1}\left[\frac{p_1}{p_2}\left(\frac{T_0}{T_1} - \frac{1}{\gamma}\right) - \left(1 - \frac{1}{\gamma}\right)\right].$$

$$\therefore \frac{T_0}{T_2} = \frac{P_2 - P_1}{\gamma P_2} + \frac{T_0 P_1}{T_1 P_2} = \frac{1}{\gamma} - \frac{P_1}{\gamma P_2} + \frac{P_1 T_0}{P_2 T_1}$$

$$\therefore \frac{T_0}{T_2} - 1 = \frac{P_1}{P_2}\left(\frac{T_0}{T_1} - \frac{1}{\gamma}\right) - \left(1 - \frac{1}{\gamma}\right)$$

$$\therefore Q = \frac{P_2 V}{\gamma - 1}\left[\frac{P_1}{P_2}\left(\frac{T_0}{T_1} - \frac{1}{\gamma}\right) - \left(1 - \frac{1}{\gamma}\right)\right]$$

Question 4.

Gas leaving the turbine of an aircraft engine flows steadily into the jet pipe at a temperature of 800˚C, a pressure of 1.8 bar, and at a velocity (relative to the jet-pipe) of 300 ms^{-1}. The gas leaves the jet pipe at 700˚C and 1.1 bar. Heat transfer is negligible. If C_p is 1.03 kJ/kg K, what is the exit velocity of the gas?

Ans. (544ms^{-1})

$$Q - W_x = \dot{M}(h_{o,out} - h_{o,in})$$

$$\therefore 0 = h_{o,out} - h_{o,in} = C_p T_{out} + \frac{1}{2}V_{out}^2 - C_p T_{in} - \frac{1}{2}V_{in}^2$$

$$\therefore V_{out} = [2C_p(T_{in} - T_{out}) + V_{in}^2]^{\frac{1}{2}}$$

$$= 40\sqrt{185} \ (m/s)$$

$$= 544.06 \ (m/s)$$

Question 5.

Air enters the compressor of a jet engine at 200 ms^{-1}, 260 K and 0.3 bar and leaves at 300 ms^{-1}, 530 K and 2.7 bar.

How much work must be supplied per kg of air if the flow is adiabatic and steady?

What is the ratio of passage area at entry to that at exit?

Ans. (298 kJ/kg:6.6)

$$(Q - W_x)_{per\ kg} = h_{o,out} - h_{o,in}$$

$$\therefore\ W_{x\ per\ kg} = C_p T_{out} + \frac{1}{2}V_{out}^2 - C_p T_{in} - \frac{1}{2}V_{in}^2$$

$$= 296350\ J/kg$$

$$= 296.35\ J/kg$$

$$\because\ M_{in} = M_{out} = M$$

$$\therefore\ M = \frac{P_{in} V_{in}}{R T_{in}} = \frac{P_{out} V_{out}}{R T_{out}}$$

$$\because\ V \propto A_{passage} V$$

$$\therefore\ \frac{A_{in}}{A_{out}} = \frac{V_{in} V_{out}}{V_{out} V_{in}} = \frac{P_{out} T_{in} V_{out}}{P_{in} T_{out} V_{in}} = 6.62$$

Question 6.

A portable high-speed drill is driven by a small turbine which uses air drawn from a high pressure air bottle of volume 0.01m^3. The bottle is rigid, and heat transfer in both bottle and turbine may be neglected.

The initial bottle pressure is 100 bar and air is used until the pressure fails to 50 bar. During the process the air in the bottle obeys $Pv^{1.4}$ = constant. The initial temperature was 300 K. The air leaves the turbine at a constant temperature of 120 K and with negligible velocity.

(a) What is the final temperature in the bottle?

(b) What is the work supplied by the turbine to the drill? **Ans.** (246 K, 70.4 kJ)

(a) $\because P_v^{1.4} = \text{constant}$

$\therefore P_{in} v_{in}^{1.4} = P_{out} v_{out}^{1.4} \quad \therefore 1\times10^7 v_{in}^{1.4} = 5\times10^6 v_{out}^{1.4}$

$\therefore 2v_{in}^{1.4} = v_{out}^{1.4} \quad \therefore 2^{\frac{5}{7}} v_{in} = v_{out}$

$\because Pv = RT \quad \therefore P_{in} v_{in} = RT_{in} ,\ P_{out} v_{out} = RT_{out}$

$$\therefore T_{out} = T_{in} \left(\frac{P_{out} v_{out}}{P_{in} v_{in}}\right)$$

$$= 300K \times \frac{5\times10^6}{1\times10^7} \times \frac{2^{\frac{5}{7}} v_{in}}{v_{in}} = 246.10K$$

(b) $\Delta E_{cv} = M_F C_F T_F - M_I C_I T_I = Q - W_x \quad \because Q = 0J$

$$\therefore W = \frac{P_I V}{RT_I} C_v T_I - \frac{P_F V}{RT_F} C_v T_F - \left(\frac{P_I V}{RT_I} - \frac{P_F V}{RT_F}\right) C_p T_{out}$$

$$= \frac{C_v V}{R}(P_I - P_F) - \left(\frac{P_I}{T_I} - \frac{P_F}{T_F}\right) \frac{C_p T_{out} V}{R}$$

$$= 70366J = 70.4kJ$$